IntelliHire	Software Requirement Specification
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Revision History

Doc.	Date	Author	Reviewer	Description of Revision
1.0	26/10/2026	Mohammad Sadman Shafiq		First issue. Requirements of the system as built and verified, traced to the BID.
1.1	27/10/2026	Mohammad Sadman Shafiq		Added requirement priorities, system diagrams, external interface requirements, data requirements, a traceability matrix, availability, portability, accessibility and backup requirements, and an approval block. Statuses updated after the end-to-end run of 27 October.

Approval

Role	Name	Signature	Date
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Contents

1. Introduction 5

1.1 Purpose 5

1.2 Scope 5

1.3 Intended Stakeholder 6

1.4 References 6

1.5 Definitions, Acronyms, and Abbreviations 6

2. Overall Description 9

2.1 Overview 9

2.2 Technical platform 11

3. Functional Requirements 13

3.1 Overview 13

3.1.1. Accounts, roles and access 14

3.1.2. Policy library and job description agent 15

3.1.3. Job posting and CV collection 17

3.1.4. CV screening and ranking (F1) 18

3.1.5. Candidate scoring (F2) 19

3.1.6. Interview scheduling 20

3.1.7. Video interview room (F3) 22

3.1.8. Real-time transcription (F4) 22

3.1.9. Interviewer co-pilot 23

3.1.10. Q&A analysis (F5) 24

3.1.11. Consolidated assessment (F6) 25

3.1.12. Candidate identification and profile (F7) 26

3.1.13. CV claim verification (F8) 28

3.1.14. Interview score (F9) 28

3.1.15. Interview quality (F10) 29

3.1.16. Candidate feedback composer (F11, innovation feature) 29

3.1.17. Candidate data deletion 32

3.1.18. Overview, system status and AI usage 33
4. User Interface 34
5. Non-Functional Requirements 37
5.1 Performance Requirements 38
5.2 Safety Requirements 39
5.3 Security Requirements 40
6. Design Constraints 43
7. Software Quality Attributes 44
8. External Interface Requirements 46
8.1 User interface rules 46
8.2 Hardware interfaces 48
8.3 Software interfaces 48
8.4 Communication interfaces 49
9. Other Requirements 50

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines what IntelliHire must do and the qualities it must have. IntelliHire is an AI-powered recruitment platform that takes a hire from the job description through CV screening and scoring, a live transcribed video interview, and an evidence-backed candidate assessment, where every AI conclusion is tied to the CV or the interview transcript.

The document turns the business needs in the Business Information Document (BID) into numbered, testable requirements. Each requirement has acceptance criteria, known limitations and the test or verification that shows it is met. It is the baseline used to build, test, review and demonstrate the system, and to trace every graded BID item (F1-F11, N1-N26) to the part of the product that delivers it.

The requirements describe the system as built and verified on 27 October 2026.

1.2 Scope

IntelliHire covers the hiring workflow from the job description to a consolidated, evidence-backed candidate assessment, in three stages.

- **Pre-interview:** job requirement profile, company policy library, AI-drafted job description, job posting and CV collection, CV upload, screening and requirement-based ranking (BID F1), criteria-weighted candidate scoring (F2), shortlist and selection, interview scheduling.
- **During the interview:** an integrated video interview room (F3), real-time speaker-attributed transcription (F4), consented candidate identification with a profile panel (F7), and an interviewer co-pilot that suggests questions and answers questions from a reference library.
- **Post-interview:** question and answer analysis with requirement coverage (F5), the consolidated assessment with export and comparison (F6), CV claim checks (F8), an interview score (F9), an interview quality review (F10), and the innovation feature, the candidate feedback composer (F11).
- **Across all stages:** role separation between interviewers and candidates, candidate data deletion, privacy controls, structured logging, health and AI usage reporting, and Docker Compose packaging.

Out of scope, as the BID section 4.2 requires: production deployment (cloud hosting, CI/CD, load balancing), enterprise authentication (SSO, LDAP), ATS or HRIS integration, payroll, offers and onboarding, multi-tenancy and billing, native mobile applications, training or fine-tuning models, and formal WCAG certification. The workflow ends at the consolidated assessment and, where a recruiter asks for it, a feedback letter to the candidate. It never produces offer, salary or onboarding content.

Job description drafting, LinkedIn posting with Google Forms collection, and interview scheduling were added as supporting features in the WBS (R-F1, R-F2, R-F5). The BID lists job posting and scheduling as out of scope for grading, so this SRS specifies them as supporting functions, not as graded core features. Voice-based identification (part of F7) is not built; identification is by face only, with the interviewer's confirmation as the fallback.

1.3 Intended Stakeholder

- **Trainers and technical mentors:** review progress against this specification and answer clarification questions.
- **Developer:** builds, tests and maintains the system against this baseline.
- **Recruiters, hiring managers and interviewers:** the system's users (the interviewer role). They define jobs, screen and score candidates, run interviews and read assessments.
- **Candidates:** join interviews through a personal link and may receive feedback. They are also the people whose data the system holds, so the privacy requirements are written for them.

1.4 References

Reference	Location
Business Information Document (BID), AI-Powered Recruitment and Interview Intelligence Platform, v1.0	docs/BID_AI_Recruitment_Platform_v2.pdf
Work Breakdown Structure and trainer review schedule	docs/IntelliHire_WBS.xlsx
System architecture (component view)	docs/IntelliHire_System_Architecture.png
System design (three-stage view)	docs/IntelliHire_System_Design.png
README (setup, usage, privacy, known limitations)	README.md
API documentation (OpenAPI, generated from the code)	http://localhost:8000/docs on a running stack
IEEE Recommended Practice for Software Requirements Specifications	IEEE Std 830-1998

1.5 Definitions, Acronyms, and Abbreviations

Term/Acronym	Definition
Job	A role being hired for, with its requirement profile and description.

Term/Acronym	Definition
Requirement profile	The job's list of requirements (skills, experience, qualifications, other criteria), each optionally marked must-have.
Must-have	A requirement a candidate must meet. A candidate with no evidence of a must-have is ruled out at screening.
User	An account in IntelliHire, with the role interviewer or candidate.
Candidate	A person who applied for a job. Stored separately from user accounts.
Screening	Checking a CV against every requirement as met, partly met or not met, each with a quote from the CV.
Evidence quote	Text an AI conclusion rests on. Code checks that it appears word for word in the CV or transcript before it counts.
Dimension	One part of a score (for example technical skills). The overall score is the weighted sum of the dimensions.
Background job	Long-running work (screening, scoring, analysis) run by the worker outside the web request, with progress reported to the screen.
Interview session	One live interview in the video room, linked to one candidate and one job.
Utterance / transcript turn	One stretch of speech by one speaker, transcribed and stored with its speaker and time.
Q&A pair	An interviewer question and the candidate's answer, as split from the transcript by the analysis.
Requirement coverage	Whether each requirement was covered, partly covered or never explored in the interview.
Consolidated assessment	The report that merges screening, scores and interview analysis into a position with evidenced findings.
Position	The assessment's computed result: strong fit, possible fit or not a fit.
Co-pilot	The interview-room assistant that suggests questions and answers the interviewer's questions.
Policy library / reference library	Company documents the job description agent may use (policies), and material the co-pilot may use for model answers (references). Kept in separate search collections.
Feedback composer	The innovation feature: drafts candidate feedback from stored evidence, checks it, and waits for a person to approve it.
Guardrail	A fixed rule or AI check a feedback letter must pass before it can be approved.
AI	Artificial intelligence.
API	Application Programming Interface.
ATS	Applicant Tracking System. "ATS-style match" means matching key terms from the job description.
BID	Business Information Document (the assignment brief).
CV	Curriculum vitae (resume).

Term/Acronym	Definition
JD	Job description.
JWT	JSON Web Token, the signed access token used for sign-in.
LangGraph	The framework used to build every AI agent as a graph of nodes with shared state.
LLM	Large language model.
MCP	Model Context Protocol. IntelliHire runs its own MCP server for Gmail, Google Calendar and SMTP.
OCR	Optical character recognition, used for scanned CVs.
RAG	Retrieval-augmented generation: the model answers from passages retrieved from a library.
SRS	Software Requirements Specification.
STT	Speech-to-text.
WBS	Work Breakdown Structure.
IH_FR / IH_NFR / IH_UI / IH_TC	Identifier prefixes in this document: functional requirement, non-functional requirement, user interface item, test case.

2. Overall Description

2.1 Overview

Product perspective

IntelliHire is a new, self-contained web application. It is not part of an existing product and does not replace an ATS. It runs entirely in Docker Compose on one machine: a FastAPI backend, a background worker, a React frontend, PostgreSQL, Redis, ChromaDB, a LiveKit video server, a transcriber service, a local Whisper speech-to-text service, MailHog for outgoing mail, and the app's own MCP server for email and calendar. It uses these external services: Groq and Google Gemini (language models, embeddings, speech-to-text), LinkedIn (job posts), Google Forms and Drive (applications), Gmail and Google Calendar (invitations), and optionally LiveKit Cloud with a Cloudflare tunnel when a candidate joins over the internet.

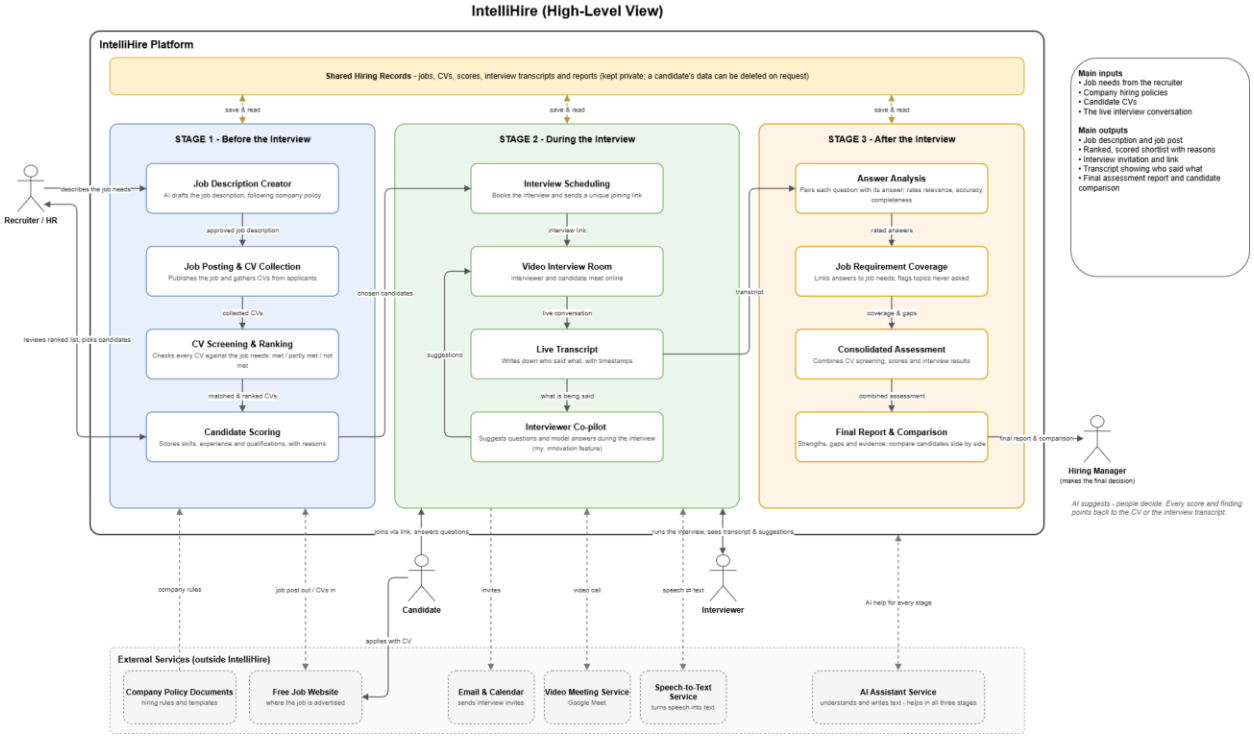


Figure 1: High-level system design across the pre-interview, during-interview and post-interview stages.

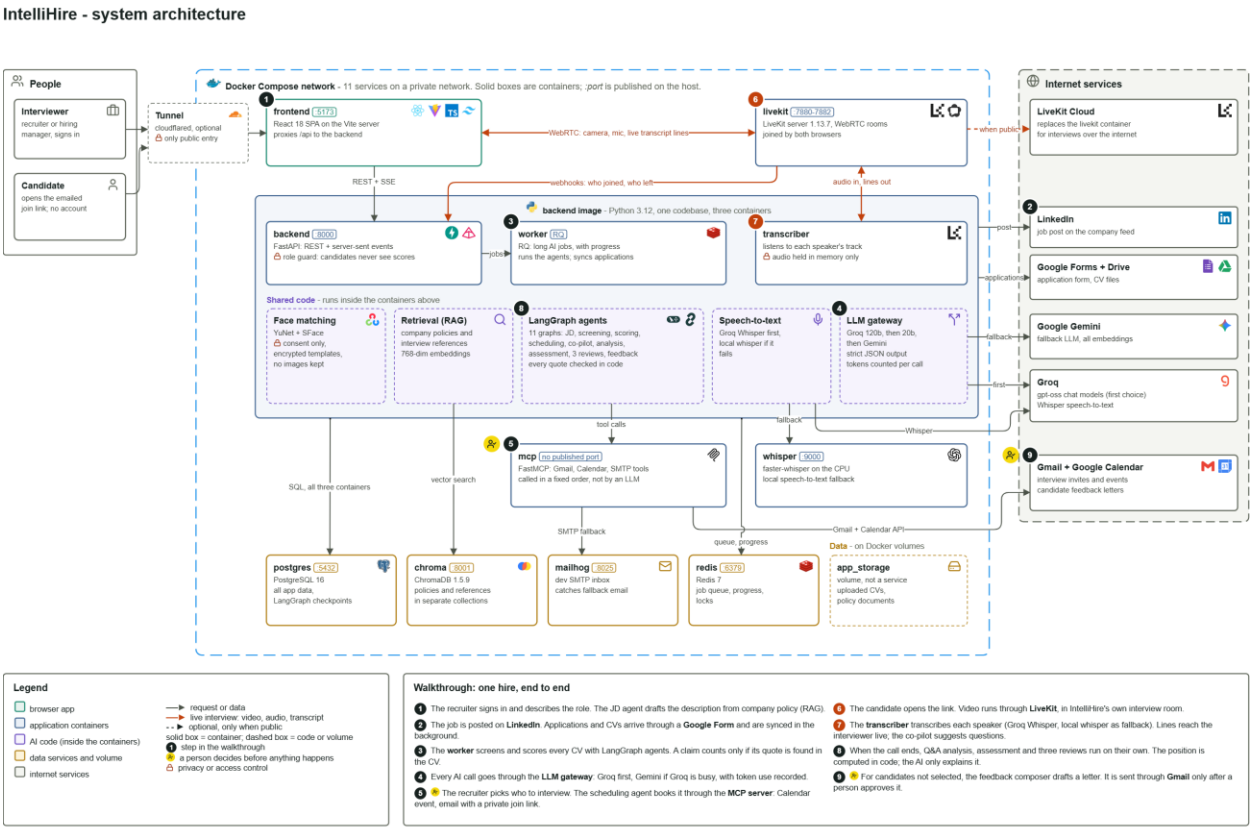


Figure 2: Component architecture as built: the eleven Compose services, the shared AI code and the internet services they call.

Product functions

IntelliHire follows a job through these steps:

- **Profile and description:** define the requirement profile and draft a policy-grounded job description.
- **Posting and applications:** post the job and collect CVs.
- **Screening and scoring:** screen and rank candidates on evidence, then score them on weighted dimensions.
- **Interviews:** shortlist and schedule, run the interview with a live transcript and a co-pilot.
- **Assessment and feedback:** analyse the interview, consolidate everything into an assessment, and optionally send feedback that a person has approved.

Every AI conclusion carries the CV text or transcript turn it rests on. Positions and numbers are computed in code; the AI explains them and never states a hiring verdict.

User characteristics

- **Interviewers and recruiters:** confident with web applications, but not technical. They need plain-language messages, visible progress for slow AI work, and evidence they can click through.
- **Candidates:** may be using the system for the first time, on any modern browser. They use only the interview page and, optionally, the identity consent. They never see scores, rankings or assessments.
- **Administrator / developer:** runs the stack with Docker commands and configures it through environment variables.

Constraints

- **The mandated stack:** Python, LangGraph, FastAPI with Pydantic, Docker Compose and GitHub (see section 6).
- **Delivery window:** 17 working days, 1 to 27 October 2026, by one developer.
- **Free tiers only:** Groq allows 1,000 requests a day and 8,000 tokens a minute per model; Gemini about 20 generation requests a day per model.
- **Personal data:** CVs, transcripts and face data must be handled as in BID section 7.3. Only synthetic or provided sample data may be used.

Assumptions and dependencies

- Docker Desktop (or Docker Engine with Compose v2), about 4 GB of free disk space and at least 8 GB of memory for Docker are available. The eleven services use about 1.3 GB of memory when idle (measured on 27 October 2026); local Whisper uses more while it transcribes. Everything runs on the CPU; no GPU is needed.
- A Google Gemini API key is provided; a Groq API key is recommended. Without Groq, Gemini answers and local Whisper transcribes, more slowly.
- For posting and collection, a LinkedIn developer app, a Google OAuth client (in Testing mode, with the user on the test-user list) and a template Google Form are set up.
- Interview participants use a Chromium-based browser or Firefox with a camera and microphone, and headphones when both are on one machine.
- The programme's 20 sample CVs were not supplied. Screening and scoring were verified on a synthetic 20-CV batch, and any batch of up to 50 CVs can take its place.

2.2 Technical platform

Component	Technology and version
Backend language and API	Python 3.12, FastAPI, Pydantic v2 (request and response validation, OpenAPI)
Agents	LangGraph with PostgreSQL checkpointing
Language models	Groq openai/gpt-oss-120b, then openai/gpt-oss-20b (strict JSON schema); fallback Google Gemini gemini-3.5-flash, gemini-3.1-flash-lite, gemini-flash-latest via google-genai

Component	Technology and version
Embeddings	Gemini gemini-embedding-001, 768 dimensions
Database	PostgreSQL 16, SQLAlchemy 2, Alembic migrations (applied when the backend starts)
Vector store	ChromaDB 1.5.9 (server and client versions matched)
Background work	Redis and RQ (one worker service)
Video	LiveKit server 1.13.7 (self-hosted), livekit 1.1.20, livekit-api 1.2.1, livekit-client 2.22.3; LiveKit Cloud for internet use
Speech-to-text	Groq Whisper whisper-large-v3-turbo; local faster-whisper as the fallback
Face matching	OpenCV 4.14 headless, YuNet detector and SFace recogniser, run locally
Email and calendar	The app's own MCP server (mcp 1.30.0, FastMCP, Streamable HTTP) with Gmail, Google Calendar and SMTP tools; MailHog locally
Frontend	React 18, Vite, TypeScript, Tailwind CSS
Authentication	Local JWT (PyJWT) with bcrypt password hashes
Packaging	Docker Compose, eleven services plus an optional tunnel (cloudflared 2026.10.1)
Development tools	Git and GitHub, VS Code, pytest, ruff, ESLint, the TypeScript compiler, Playwright for browser checks
Host operating system	Windows 11 with Docker Desktop (development); any OS that runs Docker Compose v2
Hardware	At least 8 GB of memory and 4 GB of free disk for Docker; CPU only, no GPU. Interview participants need a camera and a microphone.

3. Functional Requirements

3.1 Overview

The table lists the main functions. Each is detailed in its own subsection. Items 4, 5, 7, 8, 10 to 16 deliver BID features F1 to F11; the others support them. The Priority column follows the BID's MoSCoW ranking: the core features F1 to F6 and the innovation feature F11 are Must have, and the stretch features F7 to F10 Could have. Supporting features are Should have, except accounts and data deletion, which access separation (N9) and deletion capability (N19) make Must have. Every requirement in a subsection takes its feature's priority. Non-functional requirements apply to every feature and are not ranked separately.

Serial No	Main Features	Description	Priority
1	Accounts, roles and access	Sign-up, sign-in and the separation of interviewer and candidate capabilities.	Must have
2	Policy library and job description agent	Upload company policies; an agent drafts and refines the job description from them, with versions.	Should have
3	Job posting and CV collection	Post the job on LinkedIn and collect applications and CVs through a Google Form per job.	Should have
4	CV screening and ranking (F1)	Read CVs, check each requirement with verified quotes, rule out missing must-haves and rank candidates.	Must have (core)
5	Candidate scoring (F2)	Evidence-based dimension scores with editable weights, an aggregate score, shortlist and selection.	Must have (core)
6	Interview scheduling	Book the interview, send the invitation with a personal link, reschedule and cancel.	Should have
7	Video interview room (F3)	A browser video session between interviewer and candidate with attendance recorded.	Must have (core)
8	Real-time transcription (F4)	Speaker-attributed, timestamped transcript shown live to the interviewer and stored.	Must have (core)
9	Interviewer co-pilot	Question suggestions and a chat that answers from the CV, transcript and reference library.	Should have
10	Q&A analysis (F5)	Split the transcript into Q&A pairs, evaluate each answer and map requirement coverage.	Must have (core)

Serial No	Main Features	Description	Priority
11	Consolidated assessment (F6)	One report with a computed position, evidenced findings, export and comparison.	Must have (core)
12	Candidate identification and profile (F7)	Consented face match when the candidate joins, then a profile panel for the interviewer.	Could have (stretch)
13	CV claim verification (F8)	Compare claims in the CV with what the candidate said.	Could have (stretch)
14	Interview score (F9)	A score from the interview alone, across six dimensions.	Could have (stretch)
15	Interview quality (F10)	Review of how the interview itself was run, with recommendations.	Could have (stretch)
16	Candidate feedback composer (F11, innovation)	Evidence-based feedback letters with guardrails and human approval before sending.	Must have (innovation)
17	Candidate data deletion	Remove a candidate and everything derived from them.	Must have
18	Overview, system status and AI usage	Pipeline counts, dependency health and AI token usage.	Should have

3.1.1. Accounts, roles and access

Lets people create accounts and sign in, and keeps interviewer and candidate capabilities apart. Actors: interviewer, candidate.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_001	Users can sign up and sign in with an email address and password. Passwords are stored as bcrypt hashes and a signed JWT is issued on sign-in.	A valid sign-in returns a token. A wrong password and an unknown account return the same message.	No SSO or third-party sign-in (BID 4.2). Tokens cannot be revoked; they expire after ACCESS_TOKEN_EXPIRE_MINUTES (720).	IH_TC_001

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_002	Every account has the role interviewer or candidate. Every protected endpoint declares its required role.	A candidate token on any interviewer endpoint (screening, scoring, assessment, co-pilot) is refused with 403.	Route guards in the frontend only hide screens; the backend is the security boundary.	IH_TC_001
IH_FR_003	Self-registration can be switched off with ALLOW_REGISTRATION =false for a public address.	With registration off, sign-up is refused with a clear message.	Accounts must then be created before switching it off.	IH_TC_001
IH_FR_004	A candidate signed in to the portal sees only their own interviews and join links.	The portal shows no scores, rankings or other candidates.	The portal is a list only; candidates join through the link in their invitation.	IH_TC_001

3.1.2. Policy library and job description agent

The interviewer keeps a library of company policies and defines each job's requirement profile. A LangGraph agent drafts the job description from the profile and the policies, and refines it through chat. Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_010	The interviewer can upload policy documents (PDF, Word, text, Markdown). Text is extracted, split into passages, embedded and stored in the policy collection, with status indexed or failed with a reason.	An uploaded handbook shows indexed with its passage count, and a search such as "parental leave" returns the matching passage.	Scanned PDFs without a text layer are not OCR'd and are marked failed.	IH_TC_002

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_011	Policy search returns the passages most similar to the query and ignores passages below a minimum similarity (POLICY_MIN_SCORE, default 0.55).	Unrelated queries return no passages rather than weak matches.	The threshold is fixed per installation.	IH_TC_002
IH_FR_012	The interviewer can create and edit a job with a title, details and a requirement profile: required and preferred skills, qualifications, minimum experience and other criteria, each optionally marked must-have.	The job is saved and validated; invalid input is refused with field-level messages.	None.	IH_TC_005
IH_FR_013	Generate with AI runs the job description agent as a background job: it retrieves policy passages, drafts a structured description, checks it in code (biased wording, benefits not supported by a policy) and redrafts if needed. Progress is shown step by step.	A draft appears in roughly 20 to 40 seconds, with the policy passages it used listed under Sources. Invented benefits and biased wording are refused with the reason.	The bias check is a word list and will not catch every subtle phrasing; the interviewer reviews every draft.	IH_TC_003 IH_TC_108
IH_FR_014	The interviewer can chat with the agent to change the description or ask about it. Every change is saved as a new version; a reply that only answers changes nothing.	Each changing reply shows its version number and an Undo button. Undo restores the previous text as a new version; nothing is deleted.	The agent still refuses changes that company policy does not support.	IH_TC_004
IH_FR_015	The interviewer can edit the description by hand and restore any earlier version from History.	A manual save and a restore each create a new version.	None.	IH_TC_004

3.1.3. Job posting and CV collection

Publishes the job and collects applications and CVs. Actors: interviewer; applicants (outside the system, through LinkedIn and Google Forms).

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_020	The interviewer can connect a LinkedIn and a Google account through OAuth from Settings, Integrations. Tokens are stored and used only by the backend.	The Integrations page shows each connection's state. An expired or revoked token gives a "reconnect" message, not an error.	LinkedIn connections last 60 days with no refresh. Google refresh tokens expire after 7 days while the OAuth app is in Testing.	IH_TC_006
IH_FR_021	The job can be posted to the connected member's LinkedIn feed with an Apply link.	The post appears on LinkedIn with the job text and the form link.	LinkedIn's job-posting API is for paid partners only, so it is a feed post, not a LinkedIn Jobs listing.	IH_TC_006 IH_TC_109
IH_FR_022	A Google Form is created for each job by copying a template form. Applicants submit their details and CV through it.	A new job gets its own form link.	Applicants need a Google account to upload a file. The CV question must be added to the template by hand.	IH_TC_006
IH_FR_023	Applications are pulled into IntelliHire by Sync applications, and automatically every AUTO_SYNC_MINUTES (default 15) for jobs with a live post. CVs are copied into the app's storage and candidates are created.	A new form response appears on the Applications step after a sync. A lock keeps the automatic sync to once per interval across workers.	Not instant: there is no webhook from Google Forms.	IH_TC_007

3.1.4. CV screening and ranking (F1)

Reads every CV, checks it against the requirement profile with verified evidence, and ranks the candidates. Actor: interviewer. The screening runs as a LangGraph graph per CV, in a background job.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_030	The interviewer can upload a batch of up to 50 CVs (PDF or Word). Each file's type and size are checked before it is read (MAX_UPLOAD_MB, default 10).	Wrong types and oversized files are refused with a plain message; valid files are accepted.	Other formats are not supported.	IH_TC_008
IH_FR_031	CV text is extracted from normal, two-column and scanned (OCR) documents, and the language is detected. Unreadable or non-English CVs are set aside with a reason instead of failing the batch.	On the synthetic batch, the scan, the two-column CV and the Word file are read; the Spanish and the damaged file are set aside with reasons.	Only English CVs are screened.	IH_TC_008 IH_TC_101
IH_FR_032	Before the model sees the CV, the name, email address and phone number are removed. The screening graph extracts structured attributes (education, work history, skills, certifications, projects) and judges every requirement as met, partly met or not met by meaning, not keywords alone.	Each met or partly met result carries a quote that code finds word for word in the CV text. A result whose quote is not found counts as not met.	Indirect signals such as a university name or graduation year remain in the text.	IH_TC_009
IH_FR_033	A candidate with no evidence for a must-have requirement is ruled out and listed with the missing must-have named.	Ruled-out candidates are not ranked or scored.	None.	IH_TC_009

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_034	Candidates are ranked by weighted requirement fulfilment. Opening a candidate shows every requirement as Met, Partly met or Not met, with the CV line highlighted.	The ranking is ordered and every result links to its evidence.	None.	IH_TC_010
IH_FR_035	Screening runs as a background job, three CVs at a time, with progress per CV. A CV that fails is recorded and skipped, and pressing Screen again processes only the CVs not yet screened.	The 20-CV synthetic batch finished with 12 ranked, 6 ruled out, 2 set aside and 0 failed.	Free-tier quotas can stop a batch part-way; the next press finishes it.	IH_TC_010 IH_TC_101

3.1.5. Candidate scoring (F2)

Turns the screening evidence into dimension scores with configurable weights, a visible breakdown and a persisted result. A LangGraph scoring graph runs per candidate. Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_040	Each ranked candidate is scored on technical skills, relevant experience, qualifications and description match (an ATS-style match of key terms).	Every candidate in the screening ranking gets four dimension scores; ruled-out candidates get none.	Only candidates in the screening ranking are scored.	IH_TC_011
IH_FR_041	Each dimension starts from the verified screening evidence. The AI may move it by at most 15 points, rounded to 5, and only with a quote found word for word in the CV. The applied adjustment is stored.	An adjustment without a verified quote, or above the cap, is not applied.	The cap limits how far the model can shift a score.	IH_TC_011

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_042	On the first run for a job, the AI proposes a weight for each dimension with a reason, and the key terms for the description match. The interviewer can change the weights and save them; Ask the AI again brings back the suggestion.	Saving new weights re-ranks candidates at once with no AI call.	None.	IH_TC_012
IH_FR_043	Each candidate shows an aggregate score and its breakdown (score times weight per dimension), with a short justification and the CV lines behind each dimension.	The breakdown adds up to the aggregate shown.	None.	IH_TC_012
IH_FR_044	Scores are reproducible and persisted: temperature 0, fixed key terms, and the stored result is read back rather than recomputed. Each score records the model that produced it.	A rescore of 12 candidates gave identical scores.	An LLM is not a pure function; a fallback model may judge slightly differently.	IH_TC_011 IH_TC_101
IH_FR_045	The interviewer can add candidates to a shortlist and select a candidate for interview.	The shortlist star and the selection persist and the candidate appears on the Interviews step.	None.	IH_TC_012

3.1.6. Interview scheduling

Books interviews for selected candidates and sends the invitation. A LangGraph scheduling agent calls Gmail, Google Calendar or SMTP through the app's own MCP server, in a fixed order. Actors: interviewer; candidate (receives email).

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_050	The interviewer can schedule a selected candidate at a chosen time, or describe the wanted time ("Tuesday or Wednesday afternoon") and let the agent find a free slot on their calendar.	The event is booked and the candidate receives one email with the time in the interviewer's time zone, a personal IntelliHire link and an .ics file in UTC.	Candidates cannot pick or decline a time in the app; they reply to the email.	IH_TC_013 IH_TC_106
IH_FR_051	Email and calendar actions go only through the MCP server, called by graph nodes in a fixed order, never chosen by the model. If a later step fails, the earlier one is undone.	A failed email after a booked event removes the event.	None.	IH_TC_013
IH_FR_052	Each invitation carries a random personal link; only its hash is stored. Rescheduling sends a new time and a new link and the old link stops working. Cancelling emails the candidate first, then removes the event.	An old link after a reschedule is refused.	None.	IH_TC_014
IH_FR_053	AI-drafted emails use placeholders ({candidate_name}, {join_link}, ...) filled in by code, and code checks each draft for the content its kind must contain. No Google Meet link is ever created or sent.	The model never sees the candidate's name or address. A draft missing a required item is rejected.	None.	IH_TC_013
IH_FR_054	Without Gmail and Calendar access, the same flow sends the invitation by SMTP (MailHog locally) with no availability check, and the Interviews step says which mode is in use.	The invitation arrives in MailHog.	Gmail and Calendar need the Google connection with gmail.send, calendar.events and calendar.freebusy.	IH_TC_106

3.1.7. Video interview room (F3)

A browser-based video interview held in IntelliHire's own LiveKit room. Actors: interviewer, candidate.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_060	The interviewer opens the room from a booked interview, checks camera and microphone on a pre-join screen and starts the session. The candidate joins through their personal link once the interviewer has opened the room.	Both see and hear each other in the browser. Each participant gets a room token for their own identity.	Works on the Docker machine or a LAN by default; over the internet it needs LiveKit Cloud and the tunnel.	IH_TC_015 IH_TC_102
IH_FR_061	The call has mute, camera on or off, and end interview controls. Leaving or losing the connection does not end the interview; Rejoin returns to the same session and transcript.	A dropped connection keeps the transcript captured so far.	None.	IH_TC_015
IH_FR_062	Every session is linked to one candidate and one job. The session record keeps start and end times and participants. Join and leave are recorded from signed LiveKit webhooks, never from the browser.	The transcript page shows who was in the room and when. Unsigned webhooks are rejected.	None.	IH_TC_015
IH_FR_063	End interview ends the session for both participants and opens the transcript page for the interviewer.	The room is closed and the analysis is queued.	The call itself is not recorded.	IH_TC_015

3.1.8. Real-time transcription (F4)

A transcriber service joins each room as a hidden participant, transcribes each person's audio track, and stores a structured transcript. Actor: interviewer (reads); the system (transcribes).

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_070	Speech is cut into utterances per audio track, when the speaker pauses (SPEECH_SILENCE_MS, default 700 ms) or every 15 seconds during a long answer. The speaker is the owner of the track.	Overlapping speech gives two separate turns, each with the right speaker.	If someone uses speakers instead of headphones, echo can occasionally be transcribed twice.	IH_TC_016
IH_FR_071	Utterances are transcribed by Groq Whisper (whisper-large-v3-turbo). When Groq is rate limited, down or not set up, the local faster-whisper service takes over.	Transcription continues when Groq is unavailable.	Local Whisper on CPU falls behind when both talk continuously; the panel says it is catching up and nothing is lost.	IH_TC_016 IH_TC_027
IH_FR_072	Each turn is stored as it is transcribed, with speaker, time into the interview and clock time, and pushed live to the interviewers in the room only.	A line appears in the Live transcript after the speaker pauses: measured 0.4 to 0.75 s with Groq, 2 to 5 s with local Whisper.	The candidate never receives transcript lines.	IH_TC_016 IH_TC_102
IH_FR_073	Silence and known non-speech outputs are filtered out. Audio is held in memory only until it is transcribed and is never written to disk.	No audio files exist after an interview.	None.	IH_TC_016
IH_FR_074	After the interview the transcript page shows the full transcript with a speaker filter, search and attendance.	The stored transcript survives a restart.	None.	IH_TC_102

3.1.9. Interviewer co-pilot

Helps the interviewer during the interview with suggested questions and a chat. It reads the requirement profile, screening evidence, scores, redacted CV and live transcript. Actor: interviewer only.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_080	The interviewer can upload reference material (guides, answer keys, standards). It is indexed in its own collection, separate from policies.	The job description agent never retrieves reference passages, and the co-pilot never retrieves policies.	None.	IH_TC_017
IH_FR_081	Suggest picks the requirements most worth probing in code: screening gaps first, then must-haves, then requirements not yet discussed (embedding similarity of requirement and transcript exchanges below COPILOT_DISCUSSED_SIMILARITY, default 0.78). The model then writes a question for each.	Each question shows the requirement, why it was chosen and the CV or transcript words behind it. Quotes not found word for word are removed; questions that repeat, name an unknown requirement or touch personal characteristics are dropped.	"Not discussed yet" is a similarity judgement calibrated on one synthetic interview. Suggestions take 15 to 40 s.	IH_TC_017
IH_FR_082	Each suggestion includes what a strong answer covers, taken only from passages retrieved from the reference library. With no matching material the card says so.	Points citing a passage that was not retrieved are removed.	None.	IH_TC_017
IH_FR_083	Ask the co-pilot is a streamed chat. When the reply is complete, code checks its citations (reference passages, transcript lines, CV quotes) and saves the checked text. It declines to say whether to hire.	Unknown passage or transcript ids are removed; a CV quote not in the CV is marked "not found in the CV".	Replies start after 1 to 8 s, longer when the primary model is busy.	IH_TC_017

3.1.10. Q&A analysis (F5)

Analyses what was asked and answered, automatically after the interview ends. A LangGraph graph splits, evaluates and maps. Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_090	The analysis is queued when the interview ends and waits until the transcriber has stored the last lines.	The analysis starts by itself, about a minute after the end.	None.	IH_TC_018
IH_FR_091	The transcript is split into question and answer pairs: small talk is left out, multi-part questions are split, and a follow-up is linked to the question it follows.	Each pair references the transcript turns it was built from; turn ids that do not exist are removed by code.	None.	IH_TC_018
IH_FR_092	Each answer is scored for relevance, accuracy, completeness and articulation, with a written justification that cites transcript lines.	Clicking a line reference shows the exact words.	Scores can vary between runs (92.5 and 86.7 on one transcript); each analysis records its model, and a re-analysis replaces the previous one.	IH_TC_018
IH_FR_093	Each requirement is shown as covered, partly covered or never explored in the interview, with a follow-up question for the gaps.	Every requirement of the job appears in the coverage view.	None.	IH_TC_018

3.1.11. Consolidated assessment (F6)

Merges screening, scores and the interview analysis into one report a hiring manager can act on. Written automatically after the analysis by a LangGraph graph. Actor: interviewer / hiring manager.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_100	The assessment is written automatically once the analysis is done, from the screening results, the candidate scores and the Q&A analysis.	The report appears about half a minute after the analysis.	None.	IH_TC_019
IH_FR_101	The position (strong fit, possible fit, not a fit) is computed in code: CV fit 40%, interview answers 40%, requirement coverage 20%; strong fit from 70. A must-have with no evidence anywhere keeps the candidate below strong fit. The report shows how the number is made up.	The same inputs always give the same position. The AI explains the position and never states a hiring verdict.	The weights are fixed for every job.	IH_TC_019
IH_FR_102	The report lists strengths, weaknesses, evidence gaps and areas to verify, each with the CV line or the candidate's words behind it.	Findings whose quote or turn id cannot be verified are removed before saving.	None.	IH_TC_019
IH_FR_103	The report can be exported as a printable HTML page and as PDF.	Both exports contain the full report.	None.	IH_TC_019
IH_FR_104	The job's Assessment step lists every assessed candidate; the interviewer can select two to four and compare them side by side.	Only candidates of the same job can be compared.	None.	IH_TC_019

3.1.12. Candidate identification and profile (F7)

Checks, with the candidate's consent, that the person who joins is the candidate, then opens a profile panel for the interviewer. Actors: candidate (consents), interviewer (sees the result and decides).

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_110	Face identification is optional and needs explicit consent, given on the application form (a separate checkbox) or on the interview page, where the full wording is shown before the camera opens. Every yes, no and withdrawal is recorded with the wording version.	Without a "given" consent no face is looked at. Saying no changes nothing about the assessment.	None.	IH_TC_020
IH_FR_111	Only an encrypted face template (128 numbers from SFace) is stored, never an image. A form photo stays in Drive until an interview is booked, is then turned into a template once and deleted from Drive. A photo taken on the interview page is never saved.	The database holds no image data.	A form photo needs an optional photo question in the template form, added by hand.	IH_TC_020
IH_FR_112	When the candidate joins, their camera is checked for up to 15 seconds, one frame a second, in memory. The result is confirmed, uncertain, may be someone else or not identified. Anything short of a confident match is shown as such and the interviewer confirms or rejects; their decision wins over a later automatic check.	The match never settles identity alone below a confident result.	Thresholds are not tuned on real interviews; verified with a fake camera. Voice identification is not built.	IH_TC_020 IH_TC_110
IH_FR_113	The Profile tab opens only once identity is settled and shows the headline, experience, key skills, pre-interview score and areas to probe.	Before identity is settled the tab is closed.	None.	IH_TC_020

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_114	Templates are deleted 7 days after the interview (FACE_REFERENCE_KEEP_DAYS); unused form photos are deleted from Drive after 30 days (FACE_PHOTO_MAX_DAYS). The worker sweeps every hour. Withdrawal deletes the template at once.	Expired templates and photos are gone after the sweep.	None.	IH_TC_020

3.1.13. CV claim verification (F8)

Cross-references claims in the CV with what the candidate said. Runs automatically after the analysis. Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_120	Checkable claims are quoted from the CV and each is marked supported, answer differs from the CV, not substantiated, or never discussed, with the candidate's words and a follow-up question.	Each claim's CV quote and transcript words are verified in code.	The result is a prompt for follow-up, not a conclusion.	IH_TC_021
IH_FR_121	Claim flags are worded as prompts for a person and never accuse. Review text says "the candidate", never he or she.	Accusing or gendered words are rejected by code.	None.	IH_TC_021

3.1.14. Interview score (F9)

An interview performance score separate from the CV-based score. Runs automatically after the analysis. Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_130	The interview is scored on six dimensions: technical competency, communication, relevance of responses, problem-solving, depth of experience and alignment with the job. Each shows its weight, a measured base, the AI's small adjustment and the supporting quotes.	The score is shown beside the CV-based score, not merged into it.	None.	IH_TC_021

3.1.15. Interview quality (F10)

Reviews how the interview itself was run. Runs automatically after the analysis. Actor: interviewer, hiring manager.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_135	The review covers question effectiveness, coverage of the requirements, speaking balance, consistency with the job's other interviews, a rating for every question with a better way to ask the weak ones, and recommendations.	Each part of the review is shown on the analysis page and can be written again with its own button.	Consistency needs at least one other analysed interview for the job; with only one it is left blank.	IH_TC_021

3.1.16. Candidate feedback composer (F11, innovation feature)

Most candidates hear nothing, or a template, after a rejection. The composer turns the evidence IntelliHire already holds into a short, kind and specific letter, and keeps a person in charge of every word that is sent. It is one LangGraph graph with a stage field in its state: load_evidence, screen_inputs, select_points, draft_feedback, guardrail_check, human_review, finalize. Actors: recruiter or interviewer (decides, reviews, sends); candidate (receives).

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_140	Interview feedback can be written for an outcome the recruiter chooses (not progressing, on hold, next round). Screening feedback is only for candidates a person has marked "not selected for interview", never chosen by rank.	The AI never chooses the outcome. Unmarked candidates never appear in the screening feedback queue.	None.	IH_TC_024
IH_FR_141	Interviewer or recruiter notes are screened for protected characteristics and their proxies (age, family, origin, health, accent, "culture fit") by a rule list and then an AI check. Excluded notes are shown struck through with the reason.	A biased note never reaches the drafter. A note that cannot be checked is left out.	The screen errs towards leaving notes out.	IH_TC_022 IH_TC_023
IH_FR_142	Code selects at most a few points, each tied to a real requirement id and a stored evidence row (Q&A pair, assessment finding or screening result). The model only words these points, at temperature 0. Code writes the subject, greeting, decision sentence and sign-off.	Every sentence of the letter names the points it rests on. The Evidence panel lists them.	Screening feedback can only be as specific as the screening.	IH_TC_022

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_143	Every draft and every edit passes fixed rules (protected characteristics and proxies, scores, rankings, percentages, other candidates, dates and interview moments, personal data, promises, offer, salary or onboarding, wording that contradicts the outcome, unsupported sentences, length) and then an AI judge whose objections count only if the words it quotes are in the letter.	A failing draft is redrafted with the reasons up to twice, then the generic letter is used. A failing edit cannot be approved.	The rules and the judge are not a guarantee, which is why approval is required.	IH_TC_023
IH_FR_144	The graph stops at human_review (a LangGraph interrupt saved in the PostgreSQL checkpointer) until a person approves, edits, regenerates with an optional instruction, or discards.	Nothing is sent before approval. The run can wait for days and resume on the same thread.	None.	IH_TC_022 IH_TC_024
IH_FR_145	An approved letter is sent once, on Send, through the MCP server (Gmail when connected, otherwise SMTP). If neither is reachable or the candidate has no email address, it is saved as an .eml file in the local outbox. The name and signature are filled in only at send.	A second send of the same letter is refused. The audit trail records actions, account ids and times, not people.	The outbox is a folder; someone has to forward its letters by hand.	IH_TC_024 IH_TC_104

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_146	When the CV could not be read (under FEEDBACK_MIN_CV_CHARS, default 300) or the screening confidence is low (under FEEDBACK_MIN_CONFIDENCE, default 0.5), a polite generic letter is written without the AI. Drafting for several marked candidates runs as one background job.	The generic letter contains no specific claims.	None.	IH_TC_022 IH_TC_024

3.1.17. Candidate data deletion

Removes a candidate and everything derived from them (BID N19). Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_150	Delete candidate data (in the screening or score drawer, or DELETE /api/v1/candidates/{id}) removes, for every job: CV files and text, applications, screenings, scores, interviews with transcripts, co-pilot suggestions and chat, analyses, reviews, assessments, identity data, feedback letters with any outbox file, and the LangGraph checkpoints that hold CV text.	After deletion nothing about the candidate can be found. If the checkpoints cannot be removed, nothing is removed. Audit rows holding only ids and timings are kept.	Copies in Google Forms and Drive must be deleted there.	IH_TC_025
IH_FR_151	Every graph that sees a CV records the CV or candidate id in its run context, so deletion can find its checkpoints.	A new graph without this fails the deletion test.	None.	IH_TC_025

3.1.18. Overview, system status and AI usage

Gives the interviewer a picture of the pipeline and the health of the system. Actor: interviewer.

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_FR_160	The Overview shows a pipeline band with the count of candidates at each built stage of the pipeline, the jobs in progress and to-dos such as an expired connection.	Counts match the stored records. Only built stages are shown.	None.	IH_TC_030
IH_FR_161	Settings, System status reports the state of each dependency separately; /health and /health/ready do the same for scripts.	A stopped dependency shows as unavailable with a plain message.	None.	IH_TC_029
IH_FR_162	Every language model answer records its token counts (never its text) as an llm_calls row, with totals per agent run. A usage card on System status and GET /api/v1/usage show the totals.	Usage totals grow with each AI call.	Seen live on 27 October 2026: 58 calls and 175,226 tokens, with the same totals read back from GET /api/v1/usage.	IH_TC_030 IH_TC_112
IH_FR_163	Every long AI operation is a background job whose progress the screen polls; a finished, failed or partial job says so in plain words.	No AI request blocks the web request.	None.	IH_TC_028

4. User Interface

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
IH_UI_001	Sign in and Sign up	IH_FR_001, IH_FR_003	Email and password forms. Sign-up offers the interviewer or candidate role while registration is open.	IH_TC_001
IH_UI_002	Overview	IH_FR_160	Pipeline band with counts per stage and the jobs in progress. Main navigation: Overview, Jobs, Policies, References.	IH_TC_030
IH_UI_003	Jobs list	IH_FR_012	All jobs with their stage; New job.	IH_TC_005
IH_UI_004	Job form	IH_FR_012	Title, details and requirements, each with a Must have tick.	IH_TC_005
IH_UI_005	Job workspace (stepper)	IH_FR_012 to IH_FR_104	One page per job with the steps Profile, Description, Posting, Applications, Screening, Scoring, Interviews, Assessment, each showing its state.	IH_TC_005
IH_UI_006	Description step	IH_FR_013 to IH_FR_015	The description with Generate with AI, Edit, and a side panel with Chat, Sources and History tabs.	IH_TC_003 IH_TC_004
IH_UI_007	Posting step	IH_FR_021, IH_FR_022	Post to LinkedIn, the form link and the post state.	IH_TC_006
IH_UI_008	Applications	IH_FR_023, IH_FR_030	Received applications, Sync applications, and Add CVs (drag and drop).	IH_TC_007 IH_TC_008
IH_UI_009	Screening step and evidence drawer	IH_FR_031 to IH_FR_035	Screen N CVs with progress; ranking; ruled out by a must-have; could not be screened. The drawer shows each requirement with the CV line highlighted.	IH_TC_010

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
IH_UI_010	Scoring step and breakdown drawer	IH_FR_040 to IH_FR_045, IH_FR_140	Weights with sliders, score ranking, shortlist star, not-selected mark, Select for interview, Delete candidate data, and the Feedback review queue.	IH_TC_012 IH_TC_024
IH_UI_011	Interviews step and schedule dialog	IH_FR_050 to IH_FR_054	Schedule (pick a time or let the agent find one), Reschedule, Cancel, Start interview, and the email mode in use.	IH_TC_014
IH_UI_012	Interview room (interviewer)	IH_FR_060 to IH_FR_083, IH_FR_112, IH_FR_113	Pre-join check, the call with its controls, Live transcript, Questions and Ask the co-pilot tabs, the identity banner and the Profile tab.	IH_TC_015 IH_TC_102
IH_UI_013	Candidate interview page (/interview/:token)	IH_FR_060, IH_FR_110	Camera check, Join the interview, and the optional Confirm it's you consent card. States that the call is not recorded.	IH_TC_015 IH_TC_020
IH_UI_014	Transcript page	IH_FR_074	Full transcript with speaker filter, search and attendance.	IH_TC_102
IH_UI_015	Analysis page	IH_FR_090 to IH_FR_135	Tabs: Questions and answers, Requirement coverage, CV claims, Interview score, Interview quality.	IH_TC_018 IH_TC_021
IH_UI_016	Assessment page	IH_FR_100 to IH_FR_103, IH_FR_140 to IH_FR_146	The report with position and findings, Printable page, PDF, and the Candidate feedback tab with Guardrails, Evidence and Notes.	IH_TC_019 IH_TC_024
IH_UI_017	Assessment step and comparison	IH_FR_104	Assessed candidates; tick two to four and Compare.	IH_TC_019
IH_UI_018	Policies	IH_FR_010, IH_FR_011	Upload, status and search of company policies.	IH_TC_002
IH_UI_019	References	IH_FR_080	Upload and status of the co-pilot's reference material.	IH_TC_017

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
IH_UI_020	Settings: Integrations	IH_FR_020	Connect and reconnect LinkedIn and Google. Reached from the account block, not the main navigation.	IH_TC_006
IH_UI_021	Settings: System status	IH_FR_161, IH_FR_162	Health of each dependency and the AI usage card.	IH_TC_029 IH_TC_030
IH_UI_022	Candidate portal (My interviews)	IH_FR_004	The candidate's own interviews and join links only.	IH_TC_001

5. Non-Functional Requirements

The non-functional requirements below apply to the whole system. Each traces to the BID item in brackets.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_001	Long-running AI operations (job description, screening, scoring, scheduling, analysis, assessment, reviews, feedback) run as background jobs with progress or status feedback, never as a blocked request (N4).	Every such action returns at once with a job id and the screen shows progress.	None.	IH_TC_028
IH_NFR_002	At least one live interview runs alongside background CV processing without failure (N5).	Screening runs in the worker while an interview runs in the room and transcriber services, without failure.	Not yet verified. The end-to-end run of 27 October did not include screening while an interview was live. One worker processes background jobs.	Not run
IH_NFR_003	Failure of an external AI service gives a clear error state, never a crash or silent data loss (N6). The model chain is Groq 120b, Groq 20b, then the Gemini models; a rate-limited model cools down and is skipped.	With every model busy, the user sees a plain message and can retry; the item is recorded as not done.	Free tiers make this common under load.	IH_TC_027
IH_NFR_004	Candidate records, scores, transcripts, assessments and feedback survive an application restart (N26).	Data is present after docker compose down and up (without -v).	docker compose down -v deletes all data by design.	IH_TC_103

5.1 Performance Requirements

Response time: non-AI endpoints answer within a second on the development machine. AI work is measured per operation in the table below.

Throughput and capacity: the system is sized for one company's hiring team on one machine. It handles batches of up to 50 CVs, one or more concurrent interviews, and the post-interview chain for each interview in turn.

Degradation modes: when Groq is rate limited, the next model in the chain answers. When every model is busy, the work stops with a clear message and resumes on the next press. When Groq transcription is unavailable, local Whisper takes over at lower speed. When Gmail or Calendar is unavailable, SMTP is used.

Resource use: about 4 GB of disk for images and the Whisper model. Audio is held in memory only, and the face check reads one frame a second for at most 15 seconds.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_010	A batch of 20 CVs is parsed, screened, scored and ranked within a documented time (N1).	Measured on the 20-CV synthetic batch: about 10 s per CV, 69.5 s for the first pass; scoring 21 to 31 s per run.	Free-tier quota stopped the first pass part-way. A single-run time for the full batch is estimated at 1 to 1.5 minutes but not measured.	IH_TC_101
IH_NFR_011	Transcribed text appears with minimal perceptible delay during the session (N2).	0.4 to 0.75 s from the end of an utterance to the interviewer's panel with Groq; 2 to 5 s with local Whisper on CPU.	Groq's free tier allows 20 transcription requests a minute; above that local Whisper is used.	IH_TC_102
IH_NFR_012	The video session stays usable and stable for the length of a typical interview (N3).	Interviews held in two browsers and in Edge without dropping; Rejoin recovers a lost connection.	Internet use needs LiveKit Cloud; no TURN server is set up for the self-hosted server.	IH_TC_102
IH_NFR_013	Interactive AI features answer quickly enough to use during an interview.	Co-pilot suggestions in 15 to 40 s; chat replies start in 1 to 8 s.	Longer when the primary model is busy.	IH_TC_017

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_014	The post-interview chain completes soon after the interview ends (BID objective B5).	Analysis and assessment are ready in about one to two minutes; the three reviews follow within about a minute.	Uses three to eleven AI requests per interview on the free tier.	IH_TC_107

5.2 Safety Requirements

These requirements prevent harm to candidates from unfair, unexplained or automated decisions, and from misuse of their personal data.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_020	Scoring uses job-relevant criteria only. Age, gender, ethnicity, nationality, religion and marital status are never inputs, inferred or stored for scoring (N20). Identity fields are on a separate table and the name, email and phone are removed from the CV text before the model compares it.	The scoring prompt receives no identity fields.	Indirect signals (university, graduation year) remain; stated in the README.	IH_TC_009 IH_TC_011
IH_NFR_021	The AI supports and never replaces the human decision. Positions and component numbers are computed in code; the AI may explain them but may not state a hiring verdict. Ruled-out candidates are listed with the reason for a person to review.	No AI text in an assessment or chat states hire or reject.	None.	IH_TC_019
IH_NFR_022	Every AI conclusion is traceable to CV text or a transcript turn; unproven claims count as not met (F6).	Quotes and ids are checked in code before saving; unverifiable ones are removed.	None.	IH_TC_009 IH_TC_018 IH_TC_019

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_023	Biometric data is treated as the most sensitive category: explicit, optional consent; templates only, encrypted; short retention; never used for screening, scoring or assessment (N17).	See IH_FR_110 to IH_FR_114.	Face only; no voice biometrics.	IH_TC_020
IH_NFR_024	Nothing reaches a candidate without a person's approval: feedback letters wait for approval and are sent once; CV claim flags are prompts, never accusations or automatic rejections.	An unapproved letter cannot be sent.	None.	IH_TC_024 IH_TC_021
IH_NFR_025	The system never generates offer, salary or onboarding content.	The feedback rules reject it.	None.	IH_TC_023

5.3 Security Requirements

These requirements protect the system and candidate data from accidental or malicious access, use, change or disclosure.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_030	All keys, secrets and connection strings come from environment variables; a populated .env is never committed; .env.example holds placeholders only (N7).	The repository contains no secret.	None.	IH_TC_111
IH_NFR_031	Every API input is validated by Pydantic; uploaded files are checked for type and size before they are read (N8).	Invalid input returns a validation error in the standard envelope.	None.	IH_TC_008 IH_TC_001

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_032	Interviewer and candidate capabilities are separated in the backend: a candidate cannot reach scoring, ranking, assessment or co-pilot data (N9). Transcript lines go to interviewers only, and only the transcriber may publish data in a room.	Candidate tokens are refused on every interviewer endpoint.	None.	IH_TC_001 IH_TC_015
IH_NFR_033	Media and APIs use secure transport where exposed (N10). Public use goes over HTTPS (Cloudflare tunnel) and LiveKit Cloud; media is DTLS-SRTP encrypted.	The public address is HTTPS only.	Local signalling is plain ws:// on localhost, a documented local-development exception.	IH_TC_102
IH_NFR_034	Dependency versions are pinned exactly; no unmaintained or unnecessary packages (N11).	Every requirements and package file uses exact versions.	None.	IH_TC_111
IH_NFR_035	Error responses never contain stack traces, file paths, configuration or prompt content (N12). All errors use one envelope with a plain message.	Unexpected errors return a generic message; details go only to the redacted log.	None.	IH_TC_001
IH_NFR_036	Logs are structured and never contain names, email addresses, phone numbers, CV or transcript text, prompts or secrets (N22). Values under sensitive keys are blanked and email addresses masked.	The redaction tests pass on every log processor.	None.	IH_TC_026

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_037	Candidate links are random tokens stored only as hashes; face templates are encrypted; LiveKit webhooks are signature-checked; OAuth tokens stay in the backend and are passed to the MCP server per request, which stores none and has no published port.	A forged webhook or an old link is refused.	None.	IH_TC_014 IH_TC_015 IH_TC_020
IH_NFR_038	Only synthetic or provided sample data is used. No CVs, recordings, transcripts or database dumps are committed; those paths are git-ignored (N13, N14, N15).	sample_data/ is git-ignored; test CVs are invented.	None.	IH_TC_111

6. Design Constraints

These design decisions are mandated by the BID or were taken during development. They must be kept.

- **Python for all backend and AI logic:** version 3.12.
- **LangGraph for every agent:** each is a real multi-step graph with defined state, nodes, edges, error handling and retry paths. A single LLM call wrapped in a one-node graph is not acceptable (BID 8.2). Agents: job description, screening, scoring, scheduling, co-pilot, Q&A analysis, assessment, CV claims, interview score, interview quality, feedback composer.
- **FastAPI with Pydantic:** every backend service and automatic OpenAPI documentation.
- **Docker Compose:** the whole application starts with one documented command (docker compose up --build). Named volumes keep data; configuration comes from environment variables only.
- **GitHub:** incremental conventional commits, a working main branch, and the final submission tagged v1.0-final.
- **Layering:** endpoints hold only HTTP concerns; logic lives in services and agents; every third-party HTTP call goes through one tools module per provider.
- **AI calls:** every model and embedding call goes through one service, so the Groq then Gemini chain, cool-downs and token counting apply everywhere. Model output schemas must work in Groq's strict JSON mode.
- **Outward actions:** email and calendar go only through the app's MCP server, called by graph nodes in a fixed order, never chosen by a model.
- **Interviews are held in IntelliHire's own LiveKit room:** never Google Meet.
- **Deviations from the architecture diagram: local JWT instead of Firebase (D1); Groq Whisper and local faster-whisper instead of Google Cloud Speech (D2, D17); LiveKit instead of Meet (D17); LinkedIn feed and Google Forms for posting (D13); Groq as the default model with Gemini as the fallback (D22).**
- **Free-tier AI quotas:** these shape batch sizes and retries (see section 2.1).

7. Software Quality Attributes

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_040	Reproducibility: the application starts on a clean machine using only the documented Docker commands (N21, D6).	docker compose down -v then up recreates every table through the automatic migrations and all services report healthy.	LiveKit has no healthcheck.	IH_TC_103
IH_NFR_041	Maintainability and code quality: consistent structure, meaningful names, separation of concerns, no commented-out or dead code (N23).	ruff check, ruff format, the TypeScript type-check, ESLint and the frontend build all pass.	None.	IH_TC_105
IH_NFR_042	Testability: graph builders take their model and retrieval functions as parameters so tests inject fakes; third-party HTTP can be swapped for fake providers.	The automated suite (432 backend tests, 10 MCP server tests) passes inside the containers without calling a real model.	Live model behaviour is checked separately by live runs.	IH_TC_001 to IH_TC_031
IH_NFR_043	Auditability: every graph run writes an AgentRun row (graph, status, timings, token totals, ids) and every description change a version.	Each AI action can be traced to a run row.	Audit rows hold ids only, never people.	IH_TC_031
IH_NFR_044	Observability: structured logging for key operations and failures, a liveness and a readiness probe, and the System status page (N22).	Each dependency's state is visible without reading container logs.	None.	IH_TC_029
IH_NFR_045	Usability: the interface is usable without explanation; long operations show progress; errors are written for a non-technical reader (N25).	Every background job shows progress and every error message is plain language.	None.	IH_TC_108

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_046	Documentation: the README covers purpose, architecture, setup, configuration, usage, privacy and known limitations; endpoints are documented through OpenAPI (N24, G2).	The README and /docs are complete for every built feature.	None.	IH_TC_111
IH_NFR_047	Reliability: LangGraph checkpoints in PostgreSQL let a long run resume instead of repeating every model call; a batch skips a failing item and continues.	A failed CV does not stop its batch; an interrupted feedback run resumes on its thread.	None.	IH_TC_009 IH_TC_022
IH_NFR_048	Availability: every service, the optional tunnel included, restarts by itself after a failure (restart: unless-stopped), and the backend and its dependencies report their state through healthchecks and the readiness endpoint.	docker compose ps shows every service running and every healthcheck healthy; data is intact after a restart.	One machine, one worker, no failover (production deployment is out of scope, BID 4.2). LiveKit has no healthcheck. Recovery after a host reboot has not been tested on its own.	IH_TC_103
IH_NFR_049	Portability: the application runs on any host with Docker Compose v2. Every service is a container on an exactly pinned image, and nothing is installed on the host except Docker.	docker compose up --build starts the stack on a clean machine.	Verified only on Windows 11 with Docker Desktop. Browsers on another machine need LiveKit's node_ip set, or LiveKit Cloud.	IH_TC_103

8. External Interface Requirements

8.1 User interface rules

Interface rules that apply to every screen. The screens themselves are listed in section 4.

UI No.	UI Name	Related Info No.	Description	Notes	Test case Identifier
IH_UI_101	Progress for long work	IH_NFR_001, IH_NFR_045	Every background job shows step or item progress and its final state (done, partly done, failed).	Polled from the background job status.	IH_TC_028
IH_UI_102	Plain-language errors	IH_NFR_035, IH_NFR_045	Errors say what happened and what to do, with no technical detail.	Same envelope for every error.	IH_TC_001
IH_UI_103	Evidence highlighting	IH_NFR_022	CV quotes and transcript lines behind a conclusion are highlighted and can be opened.	Highlighter yellow is used only for evidence and the current item.	IH_TC_010
IH_UI_104	Role-based screens	IH_FR_002, IH_NFR_032	Interviewers see the full navigation; candidates see only their portal and interview page.	Screens are hidden, but the backend enforces access.	IH_TC_001
IH_UI_105	Only built features shown	IH_FR_160	The Overview and navigation list only built modules; a new module becomes a new step in the job workspace.	Main navigation: Overview, Jobs, Policies, References. Settings from the account block.	IH_TC_030
IH_UI_106	Confirmation of irreversible actions	IH_FR_145, IH_FR_150	Send, Discard and Delete candidate data ask for confirmation first.	Browser confirmation dialog.	IH_TC_024 IH_TC_025

UI No.	UI Name	Related Info No.	Description	Notes	Test case Identifier
IH_UI_107	Consistent visual language	IH_NFR_045	Shared page header, card, empty state and button components; sentence case; Schibsted Grotesk font; green-leaning neutral palette.	Design tokens in tailwind.config.js.	IH_TC_105
IH_UI_108	Accessibility	IH_NFR_045	Controls are native buttons, links and form fields that work from the keyboard; icon-only buttons and page regions carry accessible names (aria-label, aria-labelledby).	No formal WCAG audit (out of scope, BID 4.2) and no screen-reader test.	Not run

8.2 Hardware interfaces

Camera and microphone: the interview room reaches them through the browser's media permissions. The pre-join check shows the camera picture and the microphone level before joining, and headphones are advised when both participants share one machine.

Face identification reads frames from the candidate's camera only after the candidate has given consent. No other device, GPU or special hardware is used; speech-to-text and face matching run on the CPU.

8.3 Software interfaces

External systems the application exchanges data with. Every third-party HTTP call goes through one tools module per provider; email and calendar go through the application's own MCP server.

System	Used for	Interface	Data exchanged
Groq	Language model answers (openai/gpt-oss-120b, then 20b); Whisper speech-to-text	HTTPS REST, OpenAI-compatible, strict JSON schema	Prompts built from CV, transcript and job text; the audio of each utterance
Google Gemini	Fallback language models; embeddings (gemini-embedding-001, 768 dimensions)	google-genai SDK over HTTPS	Prompts; policy, reference and transcript text to embed
Google Forms and Drive	One application form per job; CV and photo files	Forms v1 and Drive v3 REST, OAuth 2.0	Job title and description out; form answers, CVs and photos in
Gmail and Google Calendar	Invitations, reschedules, cancellations, feedback letters; busy-time check	Through the MCP server; REST with OAuth 2.0 (gmail.send, calendar.events, calendar.freebusy)	Candidate name and email, interview time, join link, letter text
LinkedIn	Job post on the member feed	Posts REST API, OAuth 2.0	Job description text and the form link
LiveKit (server or Cloud)	Video rooms	Server API (livekit-api 1.2.1) for room tokens and closing rooms; signed webhooks	Room names and participant identities; media between browsers
MCP server (internal)	Email and calendar tools for the backend	MCP Streamable HTTP (mcp 1.30.0) at http://mcp:8100/mcp, not published	Tool calls from graph nodes in a fixed order
PostgreSQL 16	All records and LangGraph checkpoints	SQLAlchemy 2 over the Compose network	Every record in section 9
ChromaDB 1.5.9	Policy and reference search	HTTP client, matched to the server version	Embedded passages of policies and references

System	Used for	Interface	Data exchanged
Redis 7.4 and RQ	Background job queue and progress	Redis protocol	Job ids and progress; no candidate text
SMTP and MailHog	Fallback email and the local outbox	SMTP on 1025; MailHog web view on 8025	Invitation and feedback emails
Local Whisper service	Speech-to-text fallback	HTTP on 9000	The audio of each utterance, held in memory

8.4 Communication interfaces

Channel	Protocol and port	Notes
Browser to application	HTTP on 5173 locally; HTTPS through the Cloudflare tunnel for public use	Vite serves the page and proxies /api and /health to the backend on 8000, so one address carries the page, the API and the webhook.
REST API	JSON over HTTP with a JWT bearer token	OpenAPI documentation at /docs; one error envelope for every error.
Live updates	Server-sent events; polling	The co-pilot chat streams as server-sent events; background job progress is polled.
Video and audio	WebRTC: 7880 signalling (WebSocket), 7881/tcp and 7882/udp media; LiveKit Cloud over TLS for internet use	Transcript lines reach interviewers as LiveKit data messages; only the transcriber may publish them.
Webhooks and callbacks	HTTPS POST to /api/v1/livekit/webhook; OAuth callbacks at /api/v1/integrations/linkedin/callback and /google/callback	Webhooks are signature-checked; attendance is recorded only from them.
Email	Gmail API through the MCP server; SMTP as the fallback	Times in emails are in the interviewer's time zone; .ics files use UTC.
Internal services	The Compose network	The MCP server has no published port.

9. Other Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_050	Database: PostgreSQL holds all records; the schema is managed by Alembic migrations applied when the backend starts; LangGraph checkpoint tables are excluded from autogeneration.	alembic check reports no drift.	None.	IH_TC_103
IH_NFR_051	Third-party disclosure: the README lists which external services receive candidate data and what each receives (N18).	Groq, Gemini, Google Forms and Drive, Gmail and Calendar, LinkedIn, LiveKit Cloud and Cloudflare are each described.	The retention terms of Groq and Google govern their copies.	IH_TC_111
IH_NFR_052	Data minimisation: raw audio and video are not kept; only the transcript is stored. Face images are never stored (N16).	No audio, video or image files exist after an interview.	None.	IH_TC_016 IH_TC_020
IH_NFR_053	Language and time zones: screening supports English CVs; times in emails are in the interviewer's time zone and times in calendar files are UTC.	Non-English CVs are set aside with a reason.	Other languages are not screened.	IH_TC_008 IH_TC_013
IH_NFR_054	Budget: the system runs on free tiers (Groq, Gemini, LiveKit Cloud free plan, Cloudflare quick tunnel) with no paid service.	The full workflow ran on free tiers.	Throughput limits; paid keys remove them.	IH_TC_101

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
IH_NFR_055	Schedule and delivery: delivered within the programme window (1 to 27 October 2026) with the README, docs, incremental commit history and a final tag (G1 to G8).	The repository is shared and tagged v1.0-final.	Delivered and tagged v1.0-final on 27 October 2026. The programme's sample CVs were never supplied; the synthetic batch was used instead.	IH_TC_111 IH_TC_112
IH_NFR_056	Backup and recovery: records, files and vectors live in named Docker volumes (pg_data, app_storage, chroma_data, redis_data, whisper_models) that survive docker compose down and image rebuilds. LangGraph checkpoints let an interrupted run resume (IH_NFR_047).	Data is intact after docker compose down and up, and after the images are rebuilt with the volumes kept.	No automated backup or point-in-time recovery; a backup is a manual pg_dump plus a copy of the app_storage and chroma_data volumes. docker compose down -v deletes everything.	IH_TC_103

Data requirements

What the application stores, the personal data in it, and how long it is kept. Anything that holds candidate data is removed by Delete candidate data (IH_FR_150).

Data area	Tables and stores	Personal data held	Retention and deletion
Accounts and connections	users, integration_accounts	Interviewer and candidate account emails; bcrypt password hashes; LinkedIn and Google tokens, encrypted at rest	Kept until removed; tokens are replaced when an account is reconnected.
Jobs and policies	jobs, job_requirements, job_posts, job_description_versions, job_description_messages, company_policies; Chroma collections for policies and references	None about candidates	Kept; a description change is a new version, and undo restores an old one as a new version.
Candidates and CVs	candidates, applications, cv_documents, cv_extractions; CV files in the app_storage volume	Name, email, phone, CV file and its text	Until Delete candidate data, which also removes the files.
Screening and scoring	screenings, screening_requirement_results, candidate_scores, score_dimensions	Evidence quotes from the CV	Deleted with the candidate.
Interviews	interviews, interview_participants, transcript_turns	Schedule, a hash of the join link, attendance, transcript text	Deleted with the candidate. Audio and video are never stored.
Analysis and assessment	interview_analyses, qa_pairs, qa_requirement_links, assessments, assessment_findings, interview_reviews, copilot_suggestions, copilot_messages	Quotes from the transcript and the CV	Deleted with the candidate.

Data area	Tables and stores	Personal data held	Retention and deletion
Identity	biometric_consents, face_references, identity_checks	Consent record, encrypted face template, match results; never an image	Template deleted 7 days after the interview (FACE_REFERENCE_KEEP_DAYS); a waiting form photo is deleted from Drive once used or after 30 days (FACE_PHOTO_MAX_DAYS); both on deletion.
Feedback	candidate_feedback; outbox files when email is unreachable	Letter text	Deleted with the candidate, outbox file included.
Audit and usage	agent_runs, llm_calls, LangGraph checkpoint tables	agent_runs and llm_calls hold ids and token counts only; checkpoints hold CV and transcript text	Checkpoints are deleted with the candidate, found through the run's cv_document_id or candidate_id; audit rows are kept.

Traceability matrix

Each graded BID item, the requirements that deliver it and the tests that verify it.

BID item	Title	Requirements	Test cases
F1	CV screening and requirement-based ranking	IH_FR_030 to IH_FR_035	IH_TC_008, IH_TC_009, IH_TC_010, IH_TC_101
F2	AI-based candidate scoring	IH_FR_040 to IH_FR_045	IH_TC_011, IH_TC_012, IH_TC_101
F3	AI-powered video interview	IH_FR_060 to IH_FR_063	IH_TC_015, IH_TC_102
F4	Real-time speech recognition and transcription	IH_FR_070 to IH_FR_074	IH_TC_016, IH_TC_027, IH_TC_102
F5	Interview question and answer analysis	IH_FR_090 to IH_FR_093	IH_TC_018
F6	Consolidated candidate assessment	IH_FR_100 to IH_FR_104, IH_NFR_022	IH_TC_009, IH_TC_018, IH_TC_019
F7	Candidate identification and real-time profile (stretch)	IH_FR_110 to IH_FR_114	IH_TC_020, IH_TC_110
F8	CV claim verification through the interview (stretch)	IH_FR_120 to IH_FR_121	IH_TC_021
F9	Comprehensive interview score (stretch)	IH_FR_130	IH_TC_021

BID item	Title	Requirements	Test cases
F10	Interview quality assessment (stretch)	IH_FR_135	IH_TC_021
F11	Innovation feature: candidate feedback composer	IH_FR_140 to IH_FR_146	IH_TC_022, IH_TC_023, IH_TC_024, IH_TC_104
N1	CV parsing throughput	IH_NFR_010	IH_TC_101
N2	Transcription latency	IH_NFR_011	IH_TC_102
N3	Video session quality	IH_NFR_012	IH_TC_102
N4	API responsiveness	IH_NFR_001	IH_TC_028
N5	Concurrency	IH_NFR_002	Not run
N6	Graceful degradation	IH_NFR_003	IH_TC_027
N7	Credential management	IH_NFR_030	IH_TC_111
N8	Input validation	IH_NFR_031	IH_TC_001, IH_TC_008
N9	Access separation	IH_NFR_032	IH_TC_001, IH_TC_015
N10	Transport security	IH_NFR_033	IH_TC_102
N11	Dependency hygiene	IH_NFR_034	IH_TC_111
N12	Error disclosure	IH_NFR_035	IH_TC_001
N13	Mock or anonymised data	IH_NFR_038	IH_TC_111
N14	No personal data in the repository	IH_NFR_038	IH_TC_111
N15	Anonymised demonstration material	IH_NFR_038	IH_TC_111
N16	Data minimisation	IH_NFR_052	IH_TC_016, IH_TC_020
N17	Biometric caution	IH_NFR_023	IH_TC_020
N18	Third-party disclosure	IH_NFR_051	IH_TC_111
N19	Deletion capability	IH_FR_150 to IH_FR_151	IH_TC_025
N20	Fairness awareness	IH_NFR_020	IH_TC_009, IH_TC_011
N21	Reproducibility	IH_NFR_040	IH_TC_103
N22	Observability	IH_NFR_036, IH_NFR_044	IH_TC_026, IH_TC_029
N23	Code quality	IH_NFR_041	IH_TC_105
N24	Documentation	IH_NFR_046	IH_TC_111
N25	Usability	IH_NFR_045	IH_TC_108
N26	Data persistence	IH_NFR_004	IH_TC_103
G1 to G8	Submission and repository standards	IH_NFR_046, IH_NFR_055	IH_TC_111, IH_TC_112

Test case index

Automated tests run inside the containers (docker compose exec backend pytest). Manual checks are live runs against the running stack.

Test case identifier	Test or verification
IH_TC_001	backend/tests/test_auth_api.py, test_security.py: sign-up, sign-in, roles, the candidate boundary, the error envelope
IH_TC_002	test_policies_api.py, test_policy_search.py: policy upload, indexing and search
IH_TC_003	test_jd_agent.py: job description graph, review node and redraft loop
IH_TC_004	test_jd_chat.py: chat changes, versions, undo and restore
IH_TC_005	test_jobs_api.py: job profile create, edit and validation
IH_TC_006	test_job_posting.py, test_integrations.py: LinkedIn and Google connection, post and form creation
IH_TC_007	test_applications.py: application and CV intake, sync
IH_TC_008	test_documents.py, test_cv_text.py: upload checks, text extraction, OCR, language detection
IH_TC_009	test_screening_graph.py: redaction, quote verification, must-have filter
IH_TC_010	test_screening_api.py: batch screening, progress, ranking
IH_TC_011	test_scoring_graph.py: evidence scores, capped adjustments, description match
IH_TC_012	test_scoring_api.py: weights, aggregate, ranking, shortlist and selection
IH_TC_013	test_scheduling_graph.py, mcp_server/tests/test_server.py: slots, email checks, undo on failure, MCP tools
IH_TC_014	test_interviews_api.py: schedule, reschedule, cancel, candidate links
IH_TC_015	test_interview_room.py: session start, join, end, webhooks, room tokens
IH_TC_016	test_transcription.py: utterance cutting, Groq and Whisper fallback, filtering, saving
IH_TC_017	test_copilot.py, test_references_api.py: requirement choice, checked suggestions, chat citations
IH_TC_018	test_interview_analysis.py: Q&A split, evaluations, coverage
IH_TC_019	test_assessment.py: position and cap, finding checks, export, comparison
IH_TC_020	test_identity.py: consent, templates, decisions, profile gate, retention sweep
IH_TC_021	test_interview_reviews.py: CV claims, interview score, interview quality
IH_TC_022	test_feedback_graph.py: composer graph, point selection, interrupt and resume

Test case identifier	Test or verification
IH_TC_023	test_feedback_guardrails.py: rules and AI judge
IH_TC_024	test_feedback_api.py: not-selected mark, review actions, send once, outbox
IH_TC_025	test_candidate_deletion.py: deletion of rows, files and checkpoints
IH_TC_026	test_logging_redaction.py: log redaction
IH_TC_027	test_ai_fallback.py: model chain, cool-downs, clear failure
IH_TC_028	test_queue.py: background jobs and progress
IH_TC_029	test_health.py: liveness and readiness
IH_TC_030	test_overview.py, test_llm_usage.py: pipeline counts, token usage
IH_TC_031	test_skeleton_graph.py, test_mcp_config.py: graph runner and audit rows, MCP configuration
IH_TC_101	Manual: live screening and scoring of the 20-CV synthetic batch (job 23), including a rescore
IH_TC_102	Manual: interview in two headless Chromium browsers and in Edge; transcription latency with Groq and local Whisper
IH_TC_103	Manual: clean-machine start (docker compose down -v, then up) and restart persistence
IH_TC_104	Manual: feedback composer run with Groq through the worker, MailHog and Chrome (25 Oct)
IH_TC_105	Manual: ruff, frontend type-check, lint and build inside the containers
IH_TC_106	Manual: scheduling through the worker, MCP server and MailHog, and in a browser
IH_TC_107	Manual: analysis and assessment with a real model and in Chrome (21 Oct)
IH_TC_108	Manual: headless Chromium walkthrough of sign-up, policies, job, generate, chat, undo, history and edit
IH_TC_109	Manual: live LinkedIn post and Google Form (12 Oct)
IH_TC_110	Manual: identification with a fake camera in two browsers
IH_TC_111	Manual: repository review (.gitignore, no secrets or personal data, pinned versions, README)
IH_TC_112	Manual: end-to-end run through the UI on 27 October (job 35: description, LinkedIn post, Google Form applications, screening, scoring, Gmail and Calendar scheduling, three interviews, analysis, assessment and reviews; job 37: screening, scoring and feedback), with token usage read back from GET /api/v1/usage